CHE 313 HW # 3

9.5

2/17/26

KATERINA BOMPAI

D2. We are separating ethanol and water. All percentages are mol%. Column pressure is at 1.0 atm. VLE data are in Table 2-1. Find the q values and plot the feed lines for the following situations:

a. Feed is 60% ethanol and flashes in the column with V/F = 0.37.

b. Feed is 40% ethanol and is a two-phase mixture with liquid and vapor in equilibrium at a temperature of 84.1°C.

c. Feed is 40% ethanol and is a liquid at 20°C.

d. Feed is 40% ethanol and is a vapor at 120°C.

e. Feed is 50% ethanol and is a subcooled liquid. One mole of vapor must be condensed to heat 12 moles of feed to their boiling point.

f. Feed is 20% ethanol, and 70% is vaporized in a flash distillation system. The products of the flash system are fed separately to different feed stages. Calculate the two q values, and plot the two feed lines.

TABLE 2-1. *Vapor-liquid equilibrium data for ethanol and water at 1 atm y and x in mole fractions*

x_{Etoh}	x_w	y_{Etoh}	y_w	T, °C
0	1.0	0	1.0	100
0.019	0.981	0.170	0.830	95.5
0.0721	0.9279	0.3891	0.6109	89.0
0.0966	0.9034	0.4375	0.5625	86.7
0.1238	0.8762	0.4704	0.5296	85.3
0.1661	0.8339	0.5089	0.4911	84.1
0.2337	0.7663	0.5445	0.4555	82.7
0.2608	0.7392	0.5580	0.4420	82.3
0.3273	0.6727	0.5826	0.4174	81.5
0.3965	0.6035	0.6122	0.3878	80.7
0.5079	0.4921	0.6564	0.3436	79.8
0.5198	0.4802	0.6599	0.3401	79.7
0.5732	0.4268	0.6841	0.3159	79.3
0.6763	0.3237	0.7385	0.2615	78.74
0.7472	0.2528	0.7815	0.2185	78.41
0.8943	0.1057	0.8943	0.1057	78.15
1.00	0	1.00	0	78.30

a) $z = 0.6$, $\frac{V}{F} = 0.37$

① find q: fraction of liquid remaining

$q = \frac{L_F}{F} = 1 - \frac{V}{F} = 1 - 0.37$

$q = 0.63$

② Feed line: $y = \frac{q}{q-1}x - \frac{z}{q-1}$

$y = \frac{0.63}{0.63-1}x - \frac{0.6}{0.63-1}$

$z = 0.317$

Feed line a) $y = -1.7x + 1.62$

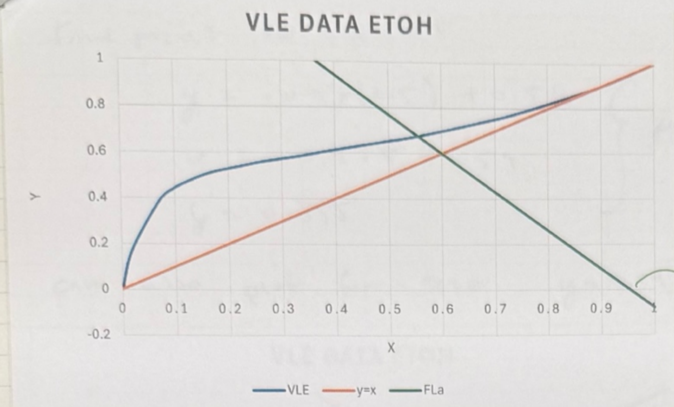
③ Plot feed line:

use 2 points or 1 point & slope

$m = \frac{\Delta y}{\Delta x} = -1.7$, ~~need~~ points $x, y < 1$ so it's easy to graph

- y-int => $0 = -1.7x + 1.62$
  $1.7x = 1.62$
  $x = \frac{1.62}{1.7} \approx 0.95$
  } point (0.95, 0)

- $z = y = x = 0.6$ } point (0.6, 0.6)



→ Feed Line a)
$y = -1.7x + 1.62$

6) ① Find $q = \frac{L}{F}$

MVC MB: $F \cdot z = Vy + Lx$

$\xrightarrow{\div F}$ $z = \frac{V}{F} y + \frac{L}{F} x$

$\frac{V}{F} = 1 - \frac{L}{F}$ $\Rightarrow$ $z = \left(1 - \frac{L}{F}\right) y + \frac{L}{F} x$

$z = y - qy + qx$

$z = y - q(y - x)$

$q(y-x) = y - z$

$q = \frac{y-z}{y-x}$

② $T = 84.1\ ^\circ C$: $y = 0.5089$, $x = 0.1651$, $z = 0.4$ } two phase stream $x < z < y$

$q = 0.32$

Feed line: $y = \frac{q}{q-1} x - \frac{z}{q-1}$

$\Rightarrow$ Feed line b: $y = \frac{0.32}{0.32-1} x - \frac{0.4}{0.32-1}$

$y = -0.47x + 0.59$

- plot:

y-int $\Rightarrow$ $0 = -0.47x + 0.59$

$x = \frac{0.59}{0.47} = 1.25$ (we'll need new point)

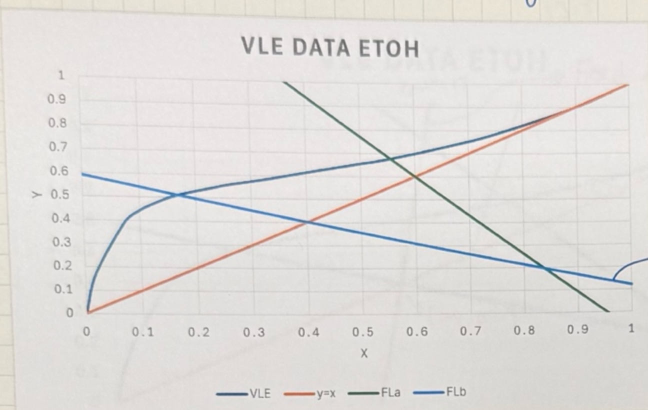
point $(0.4, 0.4)$

find point for $x = 0.5$

$y = -0.47(0.5) + 0.59$
$y = -0.235 + 0.59$
$y = 0.355$

$\}$ point $(0.5, 0.36)$

can also plot for $x=0$ $y = 0.54$ $\}$ pt $(0, 0.59)$



feed line b) $y = -0.47x + 0.59$

c) $Z = 0.4$ etOH is liquid @ $T = 20°C$

@ $T = 20°C$ etOH → subcooled liquid

① find $q = \frac{L_F}{F} = \frac{H_V - h_F}{H_V - h_L}$ >1 for subcooled liquid

use figure 2.4: wt% etOH $= \frac{Z \cdot MW_{etOH}}{Z \cdot MW_{etOH} + (1-Z) MW_{H_2O}} = \frac{0.4(46.07\frac{g}{mol})}{0.4(46.07\frac{g}{mol}) + 0.6(18.02\frac{g}{mol})} = 0.6$

$h_V = 400$

$h_L = 95$ kcal/kg

$h_F = 70$ kcal/kg

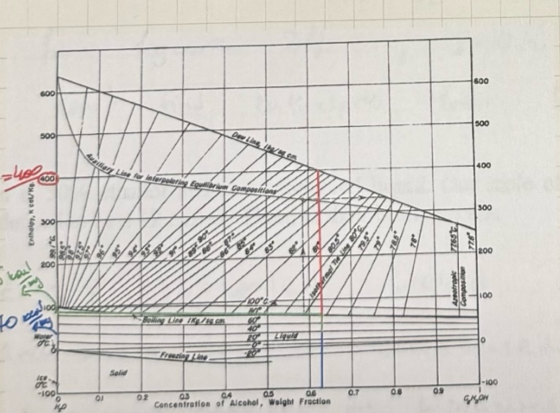


FIGURE 2-4. *Enthalpy-composition diagram for ethanol-water at a pressure of 1 kg/cm² (F. Bošnjaković Technische Thermodynamik, T. Steinkopff, Leipzig, 1935)*

$H_V = 400$ kcal/kg
$h_F = 70$ kcal/kg
$h_L = 95$ kcal/kg

$q = \frac{H_V - h_F}{H_V - h_L} = 1.08$ >1 ✓

② feed line c) $y = \frac{q}{q-1}x - \frac{z}{q-1}$

$y = 13.5x - 5$

plot: $(0.4, 0.4)$

option a) Slope $\Delta y = 13.5$

$\Delta y = 13.5\,\Delta x$, try small $\Delta x = 0.02$

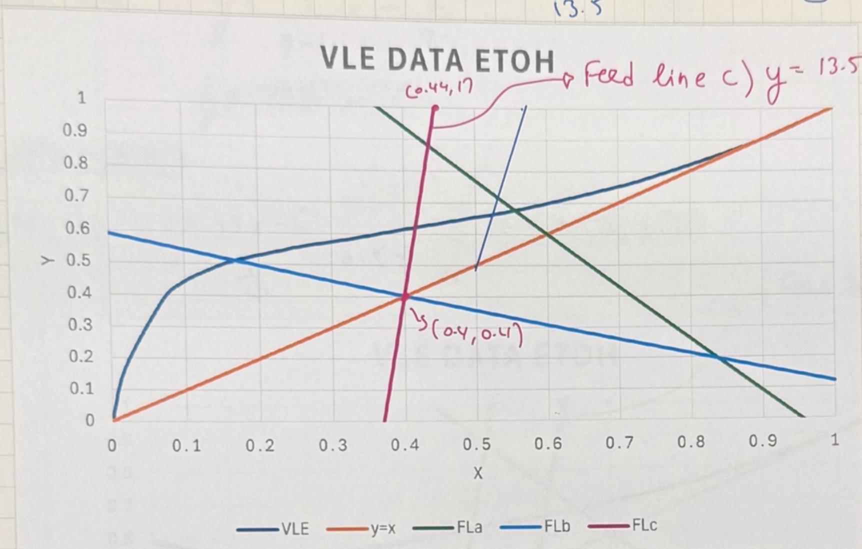
$\Delta y = 13.5(0.02) = 0.27$ change in y

So from point $(0.4, 0.4) \rightarrow (0.42, 0.67)$

q value??

option b) solve for $y=1 \Rightarrow 1 = 13.5(x) - 5$

$x = \frac{6}{13.5} = 0.44$ } pt $(0.44, 1)$



d) $z=0.4$ EtOH vapor @ 120°C → superheated vapor

expected $q<0$, slope $= \frac{q}{q-1} > 0$??

$\therefore q = -0.045$

① find $q = \frac{L_F}{F} = \frac{H_V - h_F}{H_V - h_L}$

from figure 2.4 , $z = 0.4$ wt% = 0.63

can't find enthalpies from figure at 120°C

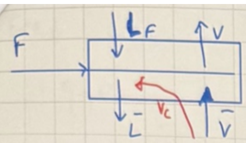
e. Feed is 50% ethanol and is a subcooled liquid. One mole of vapor must be condensed to heat 12 moles of feed to their boiling point.

$z=0.5$ subcooled liquid $q>1$

1 mol vapor must be condensed to heat 12 moles of feed to

1 mol vapor condenses → L increases

so $\frac{\text{amount condensed}}{\text{Feed}} = \frac{1}{12} \Rightarrow V_{condensed} = \left(\frac{1}{12}\right)F$



$$L_F = F + V_c$$

① find $q = \frac{L_F}{F} = \frac{F + \frac{1}{12}F}{F} = \frac{\cancel{F}(1+\frac{1}{12})}{\cancel{F}} = \frac{13}{12}$

② Feed line: $y = \frac{q}{q-1}x - \frac{z}{q-1}$

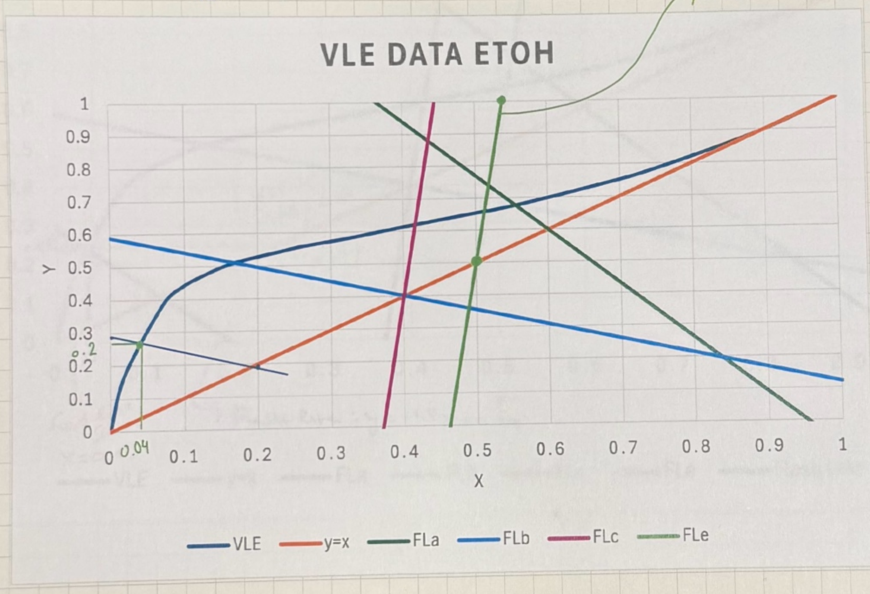
$y = 13x - 6$

pt $(0.5, 0.5)$

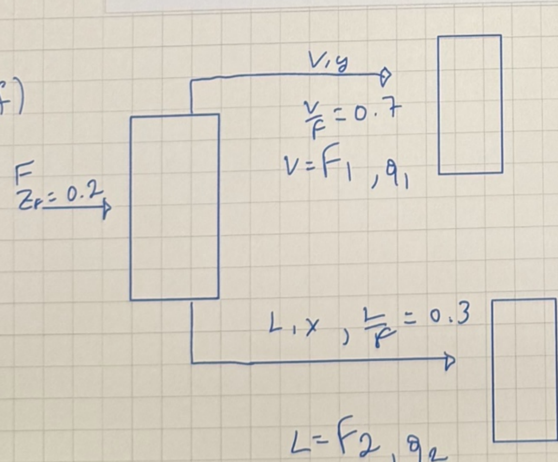
$y = 1 \Rightarrow 1 = 13x - 6$

$x = \frac{7}{13} = 0.54$ $\Big\}$ pt $(0.54, 1)$

Feed line c: $y = 13x - 6$



f)



OV MB: $F z_F = Vy + Lx$

$\xRightarrow{\div F}$ $z_F = \frac{V}{F}y + \frac{L}{F}x$

$0.2 = 0.7y + 0.3x$

$y = -\frac{0.3}{0.2}x + \frac{0.2}{0.7}$

Flash line $y = -1.5x + 0.29$

intersection Flash line $\neq$

Feed line 1) $y = \frac{q}{q-1} - \frac{z}{q-1}$

$q = 0$

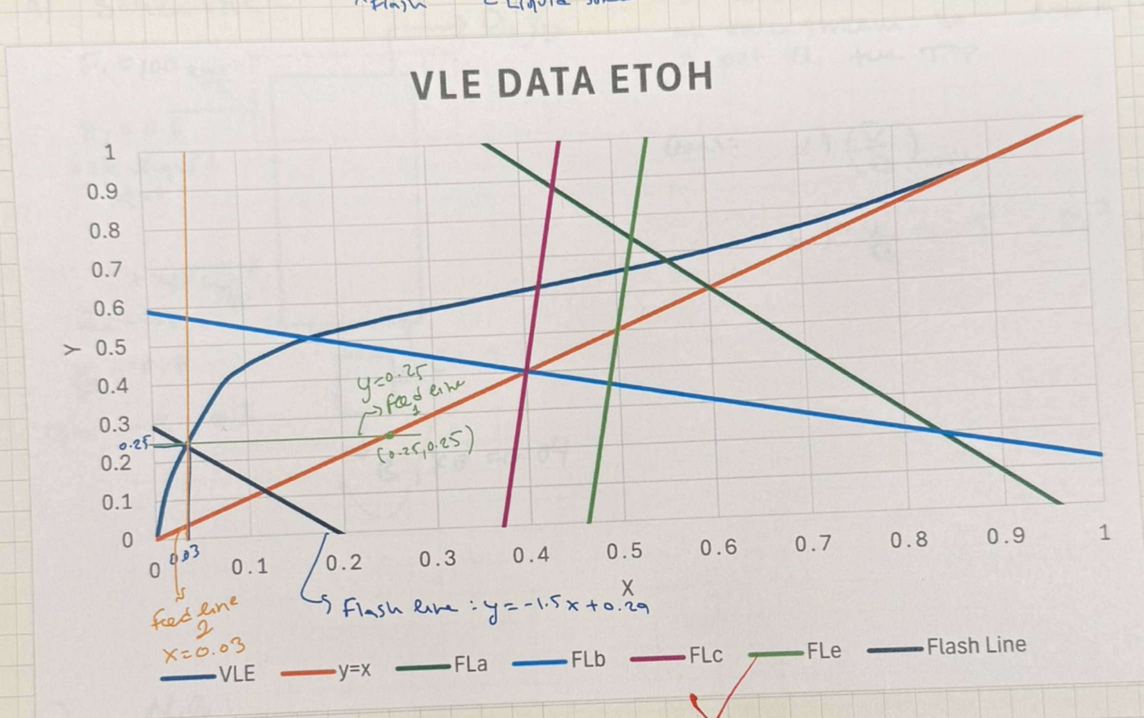
$y_{flash} = z_{vapor\ stream} = 0.25$

feed line 2) $y = \underbrace{\frac{q}{q-1}} - \frac{z}{q-1}$

$q = 1$

slope $\rightarrow \infty \rightarrow$ vertical line

$x_{flash} = z_{Liquid\ stream} = 0.03$



1.5

D4. A stripping column with two feeds is separating acetone and ethanol at 1 atm. Feed F_1 is a saturated liquid and is fed into the top of column (no condenser). Flow rate of F_1 is 100 kmol/h, and this feed is 60 mol% acetone. Feed F_2 is 40 mol% acetone, it is a two-phase feed that is 80% vapor, and flow rate is 80 kmol/h. Bottoms is 4 mol% acetone. The column has a partial reboiler. VLE data are in Problem 4.D7. → D is vapor

a. Calculate $(\bar{V}/B)_{min}$.

b. If $(\bar{V}/B) = 1.5$, find D and y_D, the optimum feed stage for feed F_2, and the total number of stages. Step off stages from the bottom upward.

a) Schematic:

$F_1 = 100\ \frac{kmol}{h}$

$z_1 = 0.6$

sat liquid

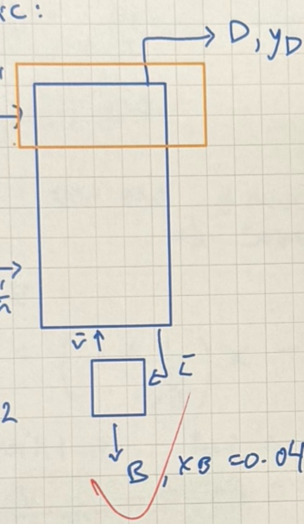
$q = 1$

$F_2 = 80\ \frac{kmol}{h}$

$z_2 = 0.4$

$\frac{V}{F} = 0.8$

$q = 1 - \frac{V}{F} = 0.2$



* we want to increase interaction of each stream so liquid stream is put at the Top

Goal: a) $\left(\frac{\bar{V}}{B}\right)_{min}$

b) $\frac{\bar{V}}{B} = 1.5$, D? y_D?

b) M.B:

overall : $F_1 + F_2 = D + B$

MVC : $F_1 z_1 + F_2 z_2 = D y_D + B x_B$

① feed line ②: $q = 0.2$, $z = 0.4$

$$y = \frac{q}{q-1}x - \frac{z}{q-1}$$

$$y_2 = -0.25\, x_1 + 0.5 \quad \text{at } (0.4, 0.4) \ \& \ (0, 0.5)$$

$x = 0$ y-int = 0.5

feed line ① sat liquid $q = 1$, $z_F = 0.6$

$y_1 = \frac{q}{q-1} x - \frac{z}{q-1}$, slope $\to \infty$, feed line 1 is vertical line at $z_F = x_F = 0.6$

③ Bottom op line:

$$y = \frac{\bar{L}}{\bar{V}} x + \left(1 - \frac{\bar{L}}{\bar{V}}\right) x_B \quad , \quad \bar{L} = \bar{V} + B \quad , \left(1 - \frac{\bar{L}}{\bar{V}}\right) = \frac{B}{\bar{V}}$$

$$\Rightarrow \frac{\bar{L}}{\bar{V}} = 1 + \frac{1}{\bar{V}/B}$$

* Feed 1 is sat liquid , $x = z_1 = 0.6$
Vapor leaving top stage must be in equilibrium w/ that liquid.

$y_D = 0.74$

$\Rightarrow$ from OV MB & MVC MB (2 eqns, 2 unknowns (D, B))
$\Rightarrow$ solve $D = 121 \frac{kmol}{h}$, $B = 59$ kmol/h

$F_1 + F_2 = D + B$
$180 = 180$ ✓

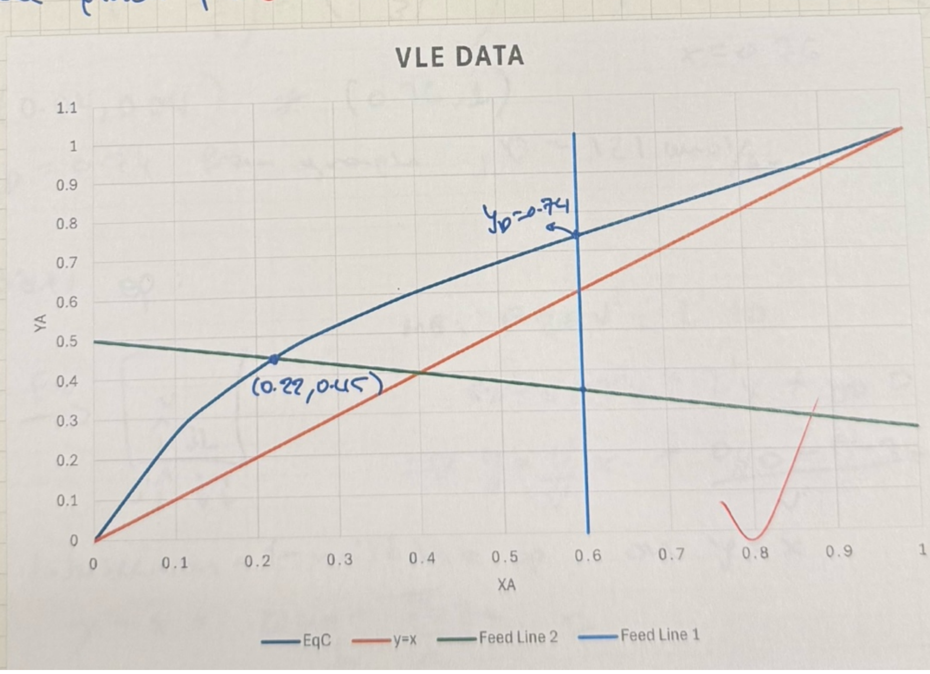
Bottom op :

$$y = \left(1 + \frac{1}{\bar{V}/B}\right) x - \left(\frac{1}{(\bar{V}/B)}\right) x_B$$

we int point of feed line 2 & EqC (0.22, 0.45)

Solving for $\frac{V}{B}_{min} = \cancel{0.78}$ 0.8113

because at minimum boil up bottom op line passes through the pinch point where it becomes feasable



Part b)

schematic:

$F_1 = 100\ \frac{kmol}{h}$

$z_1 = 0.6$

sat liquid

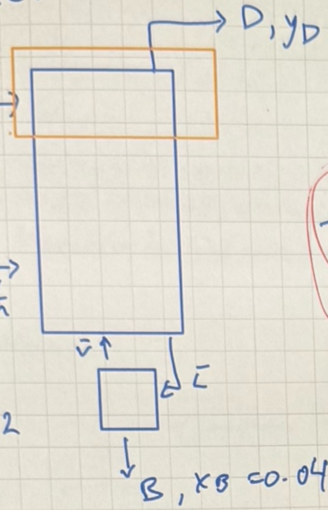
$q = 1$

$F_2 = 80\ \frac{kmol}{h}$

$z_2 = 0.4$

$\frac{V}{F} = 0.8$

$q = 1 - \frac{V}{F} = 0.2$



$\frac{\bar{V}}{B} = 1.5$

$B = \frac{\bar{V}}{1.5}$

① Bottom op line: $\bar{L} = \bar{V} + B = \bar{V} + \frac{\bar{V}}{1.5} = \frac{5}{3}\bar{V}$

$y = \frac{\bar{L}}{\bar{V}}x + \left(1 - \frac{\bar{L}}{\bar{V}}\right)x_B$

slope: $\frac{\bar{L}}{\bar{V}} = \frac{\frac{5}{3}\bar{V}}{\bar{V}} = \frac{5}{3}$

y-int: $\left(1 - \frac{\bar{L}}{\bar{V}}\right)x_B = \left(1 - \frac{5}{3}\right)0.04 = -0.0267$

$y = \frac{5}{3}x - 0.267$

$y = 1, \quad 1 = \frac{5}{3}x - 0.267$

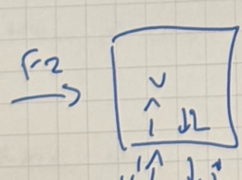
$x = \frac{1 + 0.267}{5/3}$

$x = 0.76$

pt $(0.04, 0.04)$ & $(0.76, 1)$

$y_D = 0.74$ from graph, $D = 121$ kmol/hr

middle op:



MB: $F_2 + V' = L' + D$

$F_2 z_2 + V'y = L'x + y_D \cdot D$

$\Rightarrow y = \frac{L'}{V'}x + \frac{D y_D - F_2 z_2}{V'}$

Intersection of middle op on $y = x$

$y = x = \frac{D y_D - F_2 z_2}{D - F} =$

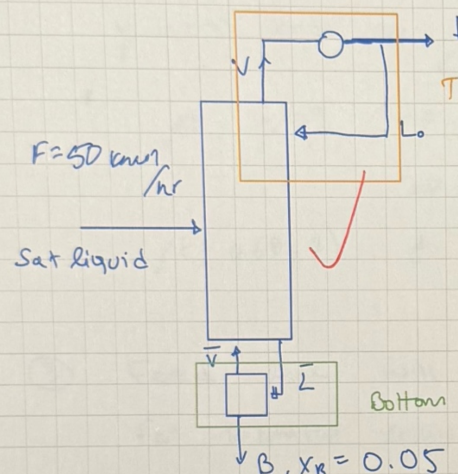
D6. A distillation column operating at 1.0 atm is separating a mixture of methanol and water. The feed is a saturated liquid. The column has a total condenser and a partial reboiler. The reflux is returned as a saturated liquid, and constant molal overflow is assumed to be valid. Feed flow rate is 50 kmol/h. We desire a bottoms composition of 5 mol% methanol and a distillate composition of 90 mol % methanol. The reflux ratio is L/D = 1, and the boilup ratio is $\bar{V}/B = 2$. VLE data are in Table 2-8.

a. Derive and plot the operating lines.

b. Starting at the top of the column, determine the optimum feed stage and the total number of equilibrium stages needed.

c. Determine the feed mole fraction methanol.

(1) Schematic:



Reflux Ratio: $\frac{L_o}{D} = 1 \Rightarrow L_o = D$

Boil up Ratio $\frac{\bar{V}}{B} = 2 \Rightarrow B = \frac{\bar{V}}{2}$

a) Equations:

Feed line: $y = \frac{q}{q-1}x - \frac{z}{q-1}$ (z_F, z_F)

Top op line: $y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$ $x_D = y$ pt$(0.9, 0.9)$

Bottom op line: $y = \frac{\bar{L}}{\bar{V}}x + (1 - \frac{\bar{L}}{\bar{V}})x_B$ $x_B = y$ pt $(0.05, 0.05)$

OVERALL MB:

$F = D + B$

TOP MB Envelope: $V = L_o + D = L_o + L_o = 2L_o$

top op line slope: $\frac{L_o}{\ } = \frac{L_o}{\ } = \frac{1}{\ } = \frac{\Delta y}{\ }$

y-int : $x=0$ $\quad y = \left(1 - \frac{L_0}{V}\right) x_D = \left(1 - \frac{1}{2}\right) 0.9$

$y_{int} = 0.45 \quad \Rightarrow y = \frac{1}{2}x + 0.45$

So for top op : pt $(0.9, 0.9)$ & pt$(0, 0.45)$

② Bottom op line:

Bottom MB envelope: $\bar{L} = \bar{V} + B = \bar{V} + \frac{\bar{V}}{2} = \frac{3}{2}\bar{V}$

Slope : $\frac{\bar{L}}{\bar{V}} = \frac{\frac{3}{2}\cancel{\bar{V}}}{\cancel{\bar{V}}} = \frac{3}{2} = \frac{\Delta y}{\Delta x}$

y-int $\Rightarrow$ $x=0$ $\Rightarrow$ $y = \left(1 - \frac{\bar{L}}{\bar{V}}\right) x_B = \left(1 - \frac{3}{2}\right) 0.05 = -0.025$

need $0 < \text{y-int} <$ to graph

pt : $y=1$ $\Rightarrow$ $1 = \frac{3}{2}x + \left(1 - \frac{3}{2}\right) 0.05$

$\Rightarrow x = 0.68$

pt$(0.68, 1)$ & pt$(0.05, 0.05)$, $y = \frac{3}{2}x - 0.025$

③ Feed line will intercept top & bottom op

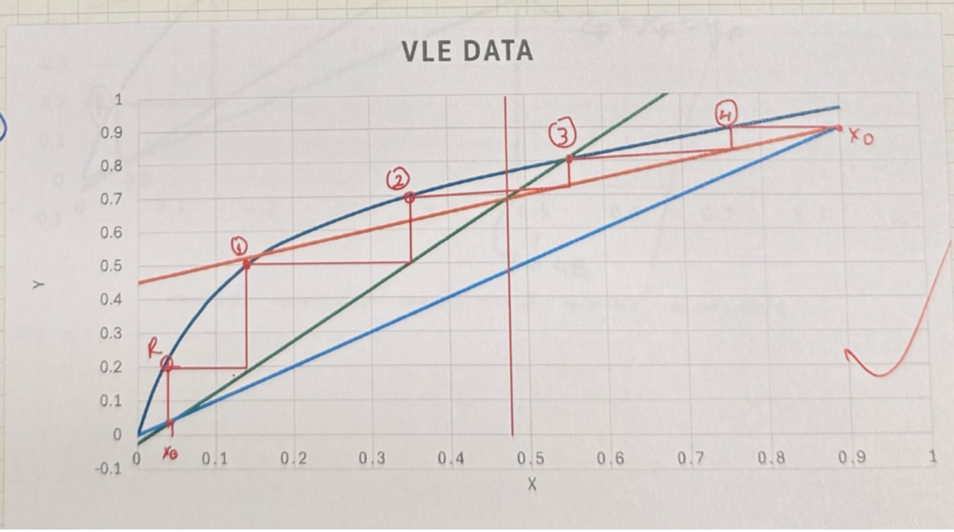
For saturated liquid feed $q=1$

$y = \frac{q}{q-1}x - \frac{z}{q-1}$ $\quad \Rightarrow$ slope $\to \infty$ $\Rightarrow$ vertical line

* The intersection of the feed line w/ $y=x$ line represents z_F because at that point vapor & liquid compositions are equal $\Rightarrow$ representing the composition of a single stream (feed).

Equilibrium Stages: 4 + reboiler

Optimum Feed Stage: between 2 & 3 so 3rd stage from top



* At Feed tray: $L_F = \bar{L} - L$

$V_F = V - \bar{V}$

$q = 1$ no vapor in feed : so vapor flow is continuous across Feed tray

$\bar{V} = V$

and from top: $V = L_0 + D$

$\Rightarrow \bar{V} = 2D$ & $\bar{V} = 2B$

$\cancel{2}B = \cancel{2}D$

$B = D$

$\Rightarrow$ MVC MB: $F z_F = D x_D + B x_B$

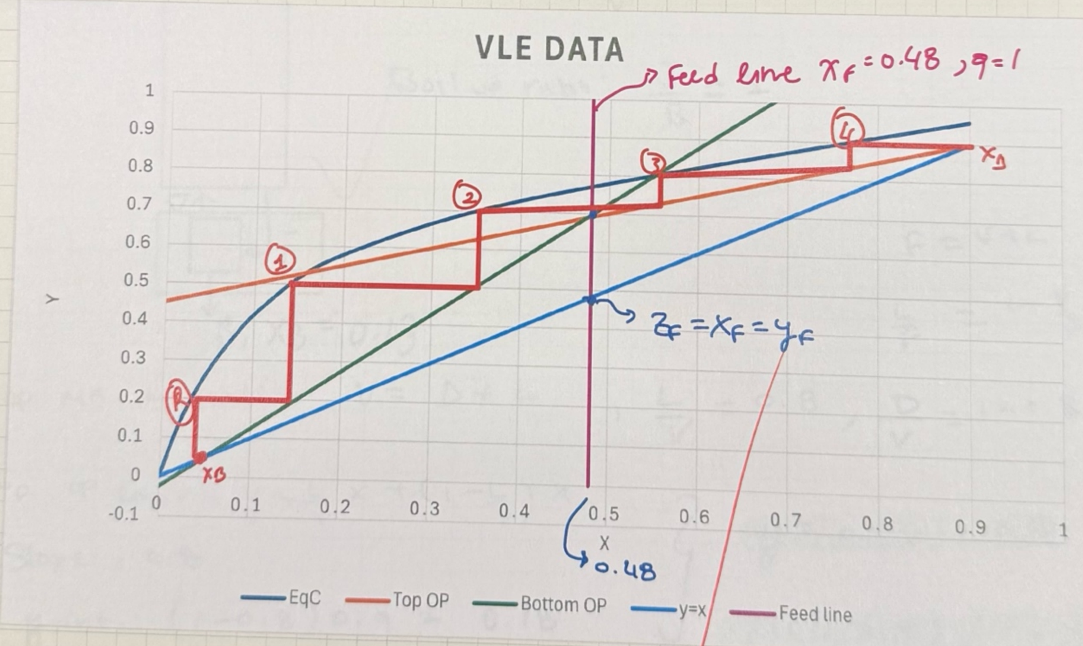
$F z_F = D(x_D + x_B)$, $F = D + B = 2D$

$2\cancel{D} \cdot z_F = \cancel{D}(x_D + x_B)$

$z_F = \dfrac{(x_D + x_B)}{2} = 0.475 \approx 0.48$

↳ $x_F = y_F$ on $y=x$ line

FINAL GRAPH:



A distillation column with a total condenser is separating a feed that is 30 mol% acetone from ethanol. Distillate $x_D = 0.90$, and bottoms $x_B = 0.13$, both mole fractions acetone. Assume CMO is valid. $L/V = 0.8$. Boilup ratio is $\bar{V}/B = 1.0$.

a. Find the composition of the liquid leaving the fifth stage below the total condenser.

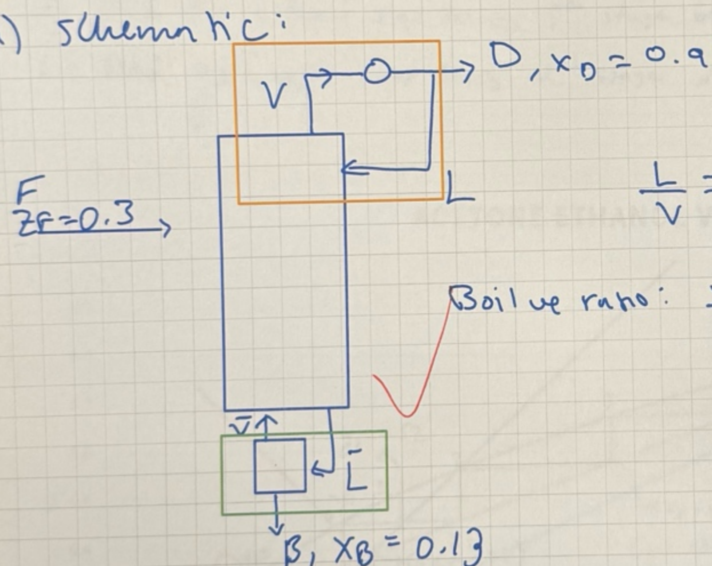
b. The column has a partial reboiler that acts as an equilibrium contact. Find the vapor composition leaving the second stage above the partial reboiler.

c. Find optimum feed plate location, total number of stages, and required q value of feed.

Equilibrium data for acetone (A) and ethanol at 1 atm (Perry et al., 1963, pp. 13–4):

x_A	0	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.50	0.60	0.70	0.80	0.90	1.0
y_A	0	0.262	0.348	0.417	0.478	0.524	0.566	0.605	0.674	0.739	0.802	0.865	0.929	1.0
T°C	78.3	-	-	-	67.3	65.9	-	63.6	-	60.4	59.1	58.0	57.0	56.1

a) schematic:



$\frac{L}{V} = 0.8$

Boilup ratio: $\frac{\bar{V}}{B} = 1$

$F = V + L$

$\frac{L}{F} = 1 - \frac{V}{F}$

① Top MB Envelope: $V = D + L$, $\frac{L}{V} = 0.8$, $\frac{D}{V} = 1 - 0.8 = 0.2$

Top op line: $y = \frac{L}{V} x + \left(1 - \frac{L}{V}\right) x_D$

Slope : 0.8

y-int : $(1 - 0.8)\,0.9 = 0.18$

$y = 0.8x + 0.18$

pt $(0, 0.18)$ & $(0.9, 0.9)$

② Bottom MB Envelope: $\bar{L} = \bar{V} + B = 2\bar{V}$, $\bar{V} = B$

Bottom op line: $y = \frac{\bar{L}}{\bar{V}} x + \left(1 - \frac{\bar{L}}{\bar{V}}\right) x_B$

slope : $\frac{\bar{L}}{\bar{V}} = \frac{2\bar{V}}{\bar{V}} = 2$

y-int : $(1-2)0.13 = -0.13$

alt pt : $y = 1 \Rightarrow 1 = 2x + (1-2)0.13$

$x = \frac{1+0.13}{2} \approx 0.565$

$y = 2x - 0.13$

pt $(0.13, 0.13)$ & $(0.57, 1)$

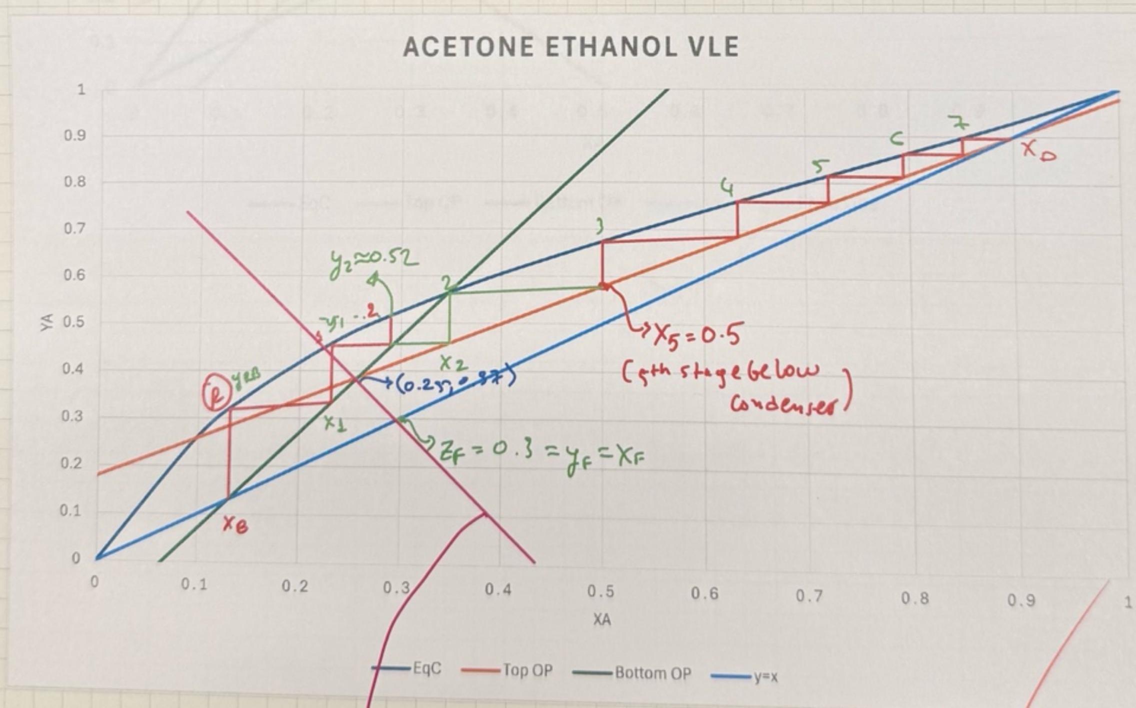
(9) feed line equation:

$y = \frac{q}{q-1}x - \frac{z}{q-1}$, $F = D + B$

$F z_F = D x_D + B x_B$

a) Find X_5 (liquid comp leaving 5^{th} stage below condenser)

b) find y_2 (vapor comp leaving 2^{nd} stage above reboiler)



feed line :

Slope $\frac{q}{q-1} = \frac{y - z_F}{x - z_F} = \frac{0.37 - 0.3}{0.25 - 0.3} = -1.4$

Slope: $\frac{q}{q-1} = -1.4 \Rightarrow q = 0.58 = \frac{L}{F}$, Feed line:

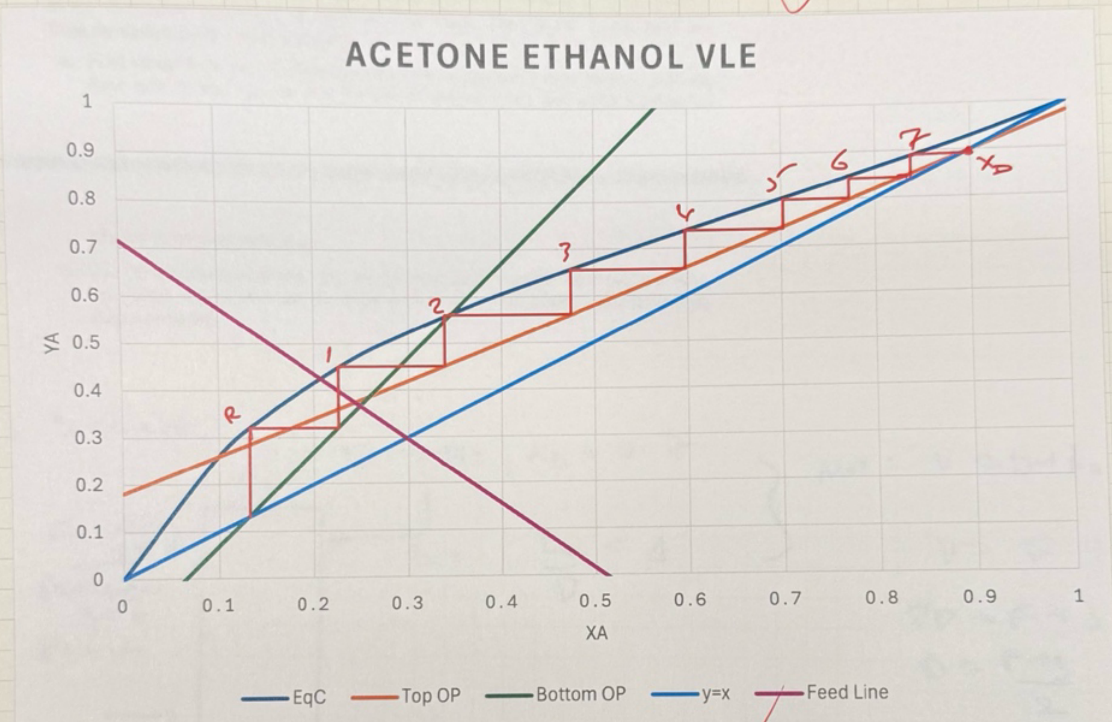
0.69

$y = -1.4x + \frac{5}{7}$

total stages: 7

optimum feed plate location : after Stage 1

FINAL GRAPH:



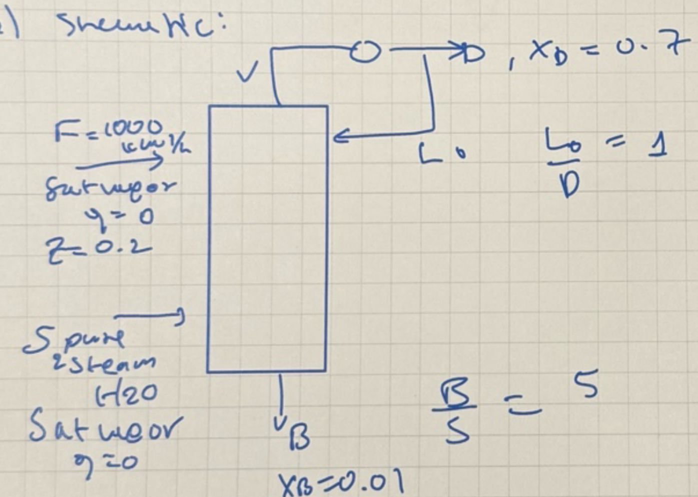
2. A distillation column is connected to another separation device. The fresh feed to the column is F = 1000.0 kmol/h of a saturated vapor that is 20.0 mol% ethanol and 80.0 mol% water. The feed is input at its optimum feed stage. The distillation column has a partial condenser and no reboiler. The column is heated with direct injection of pure steam that is a saturated vapor at 1.0 atm. The ratio bottoms flow rate/steam flow rate = B/S = 5.0. The reflux ratio $L_0/D = 1.0$. The distillate from the column is 70.0 mol% ethanol and the bottoms is 1.0 mol% ethanol. The column operates at a pressure of 1.0 atm, the reflux is returned as a saturated liquid, and CMO is assumed to be valid. VLE data are in Table 2-1.

The distillate is sent to a separation device that, because it operates on a different principle than distillation, is not limited by the presence of an azeotrope. The device produces product P and recycle stream R that is a saturated liquid returned to the distillation column at its optimum feed stage. The product P concentration from the device is 98.7 mol% ethanol.

a. Find steam flow rate S, bottoms flow rate B, distillate flow rate D, product flow rate P, and recycle rate R—all in kmol/h—and find mole fraction of ethanol in recycle stream x_R.

b. For the distillation column, find the optimum feed plate for the recycle stream, the optimum feed plate for the fresh feed, and the total number of equilibrium stages required.

a) Schematic:



MB: $V = D + L_0 = 2D$

$V = F + S$

$2D = F + S$

$D = \frac{F+S}{2}$

overall MB

$F + S = B + P$

& $\frac{B}{S} = 5$

$S = 201.9$ kmol/hr

$B = 1009.44$ kmol/hr

$P = 192.41$ kmol/hr

$\Rightarrow D = \frac{F+S}{2} = 600.95$ kmol/

Separator MB: $D = P + R \Rightarrow$

$R = 408.54$ kmol/hr

Ethanol MB: $0.7 D = 0.987 P + x_R \cdot R \Leftrightarrow x_R = 0.56$